Synoptic Question 29: The Physics of a CT Scanner

Reading Passage

Read the following article and then answer the questions that follow.

Computed Tomography (CT) is an advanced medical imaging technique that uses X-rays to create detailed cross-sectional images ("slices") of the body. Unlike a conventional X-ray which produces a 2D projection, a CT scanner acquires multiple projections from different angles.

The core components are an X-ray tube and an array of detectors mounted on a rotating gantry. The patient lies on a table that moves through the centre of the gantry. As the gantry rotates, the X-ray tube emits a fan-shaped beam that passes through the patient. The detectors on the opposite side measure the intensity of the X-rays that are transmitted.

Different tissues attenuate (absorb and scatter) X-rays to different extents. Dense materials like bone absorb more X-rays (higher attenuation) than soft tissues or air. The amount of attenuation depends on the tissue's density, atomic number, and the energy of the X-ray photons. The intensity I of an X-ray beam after passing through a thickness x of material is given by $I = I_0 e^{-\mu x}$, where I_0 is the initial intensity and μ is the linear attenuation coefficient.

A computer algorithm then processes the intensity measurements from all angles to reconstruct a cross-sectional image, assigning different shades of grey based on the calculated attenuation values for each small volume element (voxel) within the slice. Generating the high voltage needed for the X-ray tube and managing the heat produced are significant engineering challenges. The rapid rotation of the heavy gantry also involves considerable forces.

Data:

Data for a typical medical CT scanner:

Table 1: Data for CT Scanner Analysis

Parameter	**Value**
X-ray tube accelerating voltage, V_{acc}	120 kV
X-ray tube current, I_{tube}	250 mA
Effective mass of rotating gantry assembly, m_g	1500 kg
Radius of X-ray tube path during rotation, r	0.75 m
Gantry rotation period, T	0.50 s
Linear attenuation coefficient of bone, μ_{bone}	0.53 cm^{-1}
Linear attenuation coefficient of muscle, μ_{muscle}	0.19 cm^{-1}
Mass of rotating anode target, m_{anode}	1.8 kg
Specific heat capacity of anode material, c_{anode}	130 J kg^{-1} K^{-1}
Efficiency of X-ray production (electrical to X-ray)	$\approx 1\%$
Planck's constant, h	6.63×10^{-34} J s
Charge of electron, e	1.60×10^{-19} C
Speed of light, c	3.00×10^{8} m s^{-1}

Questions

(a) The X-ray tube generates X-rays by accelerating electrons.

(i) Calculate the maximum kinetic energy (in joules) gained by an electron accelerated through 120 kV.

(ii) Calculate the minimum wavelength of the X-rays produced (Bremsstrahlung radiation).

(iii) Calculate the total electrical power input to the X-ray tube.

(b) Most of the energy input to the X-ray tube is converted to heat in the anode target.

(i) Estimate the rate at which thermal energy is generated at the anode.

(ii) If this energy were absorbed by the anode for 0.20 s without any cooling, calculate the temperature rise of the anode.

(iii) Explain why the anode is usually rotated at high speed.

(c) The gantry assembly rotates rapidly around the patient.

(i) Calculate the angular velocity ω of the gantry in rad s^{-1}.

(ii) Calculate the linear speed v of the X-ray tube.

(iii) Calculate the magnitude of the centripetal force required to keep the X-ray tube moving in its circular path. (Assume its mass is 50 kg).

(d) X-rays are attenuated as they pass through the patient's body.

(i) Calculate the thickness of bone required to reduce the intensity of the X-ray beam to 10% of its incident value.

(ii) Calculate the percentage of X-ray intensity transmitted through 15 cm of muscle.

(iii) Explain how the difference in attenuation between bone and muscle allows a CT image to distinguish these tissues.

(e) The high voltage for the X-ray tube is often produced using a step-up transformer and rectifier circuit.

(i) If the primary coil of the transformer has 200 turns and is supplied with 480 V AC, calculate the number of turns required on the secondary coil to produce a peak voltage of 120 kV AC.

(ii) Briefly describe the function of a rectifier circuit in this context.

Solutions

(a) (i) Max KE $= eV_{acc} = (1.60 \times 10^{-19})(120 \times 10^3) = 1.92 \times 10^{-14}$ J.

(ii) Max photon energy E_{max} = Max electron KE.
$E_{max} = hf_{max} = \frac{hc}{\lambda_{min}}$
$\lambda_{min} = \frac{hc}{E_{max}} = \frac{(6.63\times10^{-34})(3.00\times10^8)}{1.92\times10^{-14}} = 1.036 \times 10^{-11}$ m (or 0.0104 nm).

(iii) $P_{elec} = V_{acc} \times I_{tube} = (120 \times 10^3)(250 \times 10^{-3}) = 30000$ W (or 30 kW).

(b) (i) If only 1
Rate of heat generation $P_{heat} = 0.99 \times P_{elec} = 0.99 \times 30000 = 29700$ W.

(ii) Energy deposited $Q = P_{heat} \times t = (29700)(0.20) = 5940$ J.
$Q = m_{anode}c_{anode}\Delta T$
$\Delta T = \frac{Q}{m_{anode}c_{anode}} = \frac{5940}{(1.8)(130)} = 25.4$ K (or 25.4 °C).

(iii) Rotating the anode spreads the heat generated by the electron beam over a larger area (the circumference of the focal track), preventing any single spot from overheating and melting.

(c) (i) $\omega = \frac{2\pi}{T} = \frac{2\pi}{0.50} = 12.57$ rad s^{-1} ≈ 12.6 rad s^{-1}.

(ii) $v = r\omega = (0.75)(12.57) = 9.43$ m s^{-1}.

(iii) Assuming mass of X-ray tube $m_{tube} = 50$ kg.
$F_c = m_{tube}r\omega^2 = (50)(0.75)(12.57)^2 = 5925$ N ≈ 5930 N.
(Alternatively: $F_c = m_{tube}v^2/r = (50)(9.43)^2/0.75 = 5926$ N).

(d) $\mu_{bone} = 0.53$ cm^{-1}, $\mu_{muscle} = 0.19$ cm^{-1}.

(i) We want $I = 0.10I_0$.
$0.10I_0 = I_0e^{-\mu_{bone}x}$
$0.10 = e^{-0.53x}$
$\ln(0.10) = -0.53x$
$-2.303 = -0.53x \implies x = \frac{2.303}{0.53} = 4.35$ cm.

(ii) $x = 15$ cm.
$I = I_0e^{-\mu_{muscle}x} = I_0e^{-(0.19)(15)} = I_0e^{-2.85} = I_0 \times 0.0578$.
Percentage transmitted = 5.78%.

(iii) Bone attenuates X-rays much more strongly than muscle (higher μ). In the CT image, regions corresponding to bone will appear brighter (as less X-ray intensity reaches the detector, the reconstructed value is higher/whiter), while muscle will appear darker grey. This difference in attenuation creates contrast between the tissues.

(e) (i) For an ideal transformer: $\frac{V_s}{V_p} = \frac{N_s}{N_p}$
$N_s = N_p\frac{V_s}{V_p} = (200)\frac{120\times10^3}{480} = 50000$ turns.

(ii) A rectifier circuit (typically using diodes) converts the alternating current (AC) output of the transformer's secondary coil into direct current (DC), which is required to create the constant potential difference needed to accelerate electrons in the X-ray tube.

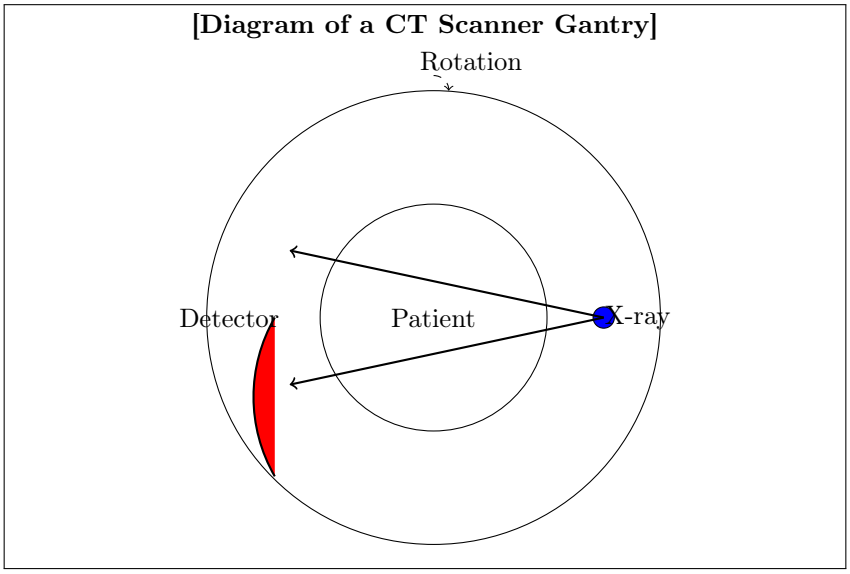

Figure 1: Simplified schematic of CT gantry components.

CIE 9702 Physics (A Level) – Paper 4 Style

Synoptic Question: CT Scanner Physics – Variant 2 (Higher Difficulty)

Instructions:

- Answer all questions.
- Give all working clearly and quote answers to an appropriate number of significant figures.
- Where appropriate, include units.

Data

Parameter	Value
Accelerating voltage of X-ray tube, V_{acc}	$1.40 \times 10^2\,\text{kV}$
Tube current, I_{tube}	$3.20 \times 10^2\,\text{mA}$
X-ray production efficiency	1.2%
Mass of rotating gantry assembly, m_g	$1.800 \times 10^3\,\text{kg}$
Radius of X-ray tube path, r	$8.0 \times 10^{-1}\,\text{m}$
Gantry rotation period, T	$4.2 \times 10^{-1}\,\text{s}$
Linear attenuation coefficient of cortical bone, μ_b	$6.2 \times 10^{-1}\,\text{cm}^{-1}$
Linear attenuation coefficient of soft tissue, μ_s	$2.1 \times 10^{-1}\,\text{cm}^{-1}$
Linear attenuation coefficient of iodine contrast region, μ_i	$3.8 \times 10^{-1}\,\text{cm}^{-1}$
Mass of anode, m_a	$2.2\,\text{kg}$
Specific heat capacity of anode, c_a	$1.40 \times 10^2\,\text{J kg}^{-1}\,\text{K}^{-1}$
Planck constant, h	$6.63 \times 10^{-34}\,\text{J s}$
Charge on electron, e	$1.60 \times 10^{-19}\,\text{C}$
Speed of light, c	$3.00 \times 10^8\,\text{m s}^{-1}$

Questions (Variant 2 only)

(a) The tube accelerates electrons through $1.40 \times 10^2\,\text{kV}$.

(i) Calculate the maximum kinetic energy of one electron in joules. [2]

(ii) Determine the minimum wavelength of emitted X-rays. [3]

(iii) Explain why the actual X-ray spectrum from the tube is continuous, even though there is a minimum wavelength. [2]

(b) The CT tube operates at $3.20 \times 10^2\,\text{mA}$ and $1.40 \times 10^2\,\text{kV}$.

(i) Calculate the electrical power input to the tube. [2]

(ii) Estimate the rate of thermal energy generation in the anode. [2]

(iii) If no cooling occurs for $1.5 \times 10^{-1}\,\text{s}$, calculate the anode temperature rise. [2]

(iv) State and explain two engineering design features that reduce thermal damage in modern CT tubes. [4]

(c) The gantry rotates with period 4.2×10^{-1} s.

(i) Calculate the angular velocity and linear speed of the X-ray tube path. [3]

(ii) Calculate the centripetal force required for a 5.5×10^{1} kg tube assembly at radius 8.0×10^{-1} m. [2]

(iii) Open-ended: discuss two physical consequences for image quality and two safety/engineering consequences if rotation period is reduced substantially (e.g. to 2.5×10^{-1} s) while other settings are unchanged. [4]

(d) A narrow beam passes through a 5.0 cm bone layer followed by 1.00×10^{1} cm soft tissue.

(i) Show that $I = I_0 e^{-(\mu_b x_b + \mu_s x_s)}$ for this two-material path. [2]

(ii) Calculate I/I_0 for the given path. [2]

(iii) The beam then crosses an additional 2.0 cm iodine-contrast vessel. Calculate the new transmitted fraction. [2]

(iv) Explain, in terms of attenuation physics and reconstruction, why contrast agents improve the visibility of blood vessels. [3]

(e) The high voltage supply uses a transformer and full-wave rectification.

(i) A primary coil has 240 turns and receives 6.00×10^{2} V RMS. Calculate the number of turns needed on the secondary to produce 1.40×10^{2} kV peak. [3]

(ii) Explain why rectification and smoothing are required for stable CT operation. [2]

(iii) Open-ended: suggest one limitation of using a simple ideal-transformer model in this application and explain its practical effect. [2]

[Total: 40 marks]

Sample Solutions and Mark Scheme (Variant 2)

Notes for examiners:

- Allow ecf (error carried forward) where appropriate.
- Credit any valid physics not explicitly listed.
- For open-ended parts, award marks by point scheme up to stated maximum.

(a) (i) $E_{\max} = eV = (1.60 \times 10^{-19})(140 \times 10^3) = \boxed{2.24 \times 10^{-14}\,\mathrm{J}}$. [2]

(ii) $E_{\max} = hc/\lambda_{\min}$,

$$\lambda_{\min} = \frac{hc}{E_{\max}} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{2.24 \times 10^{-14}} = \boxed{8.88 \times 10^{-12}\,\mathrm{m}}$$

(or $\boxed{0.00888\,\mathrm{nm}}$). [3]

(iii) Continuous spectrum explanation (any two): electrons lose variable fractions of KE in deceleration events; multiple collision histories in target; superposition of many photon energies above $\lambda_{\min}$. [2]

(b) (i) $P_{\text{in}} = VI = (140 \times 10^3)(0.320) = \boxed{4.48 \times 10^4\,\mathrm{W}}$ (44.8 kW). [2]

(ii) $P_X = 0.012P_{\text{in}}$ so $P_{\text{heat}} = 0.988P_{\text{in}}$

$$P_{\text{heat}} = 0.988(4.48 \times 10^4) = \boxed{4.43 \times 10^4\,\mathrm{W}}$$

[2]

(iii) $Q = P_{\text{heat}}t = (4.43 \times 10^4)(0.15) = 6.64 \times 10^3\,\mathrm{J}$,

$$\Delta T = \frac{Q}{m_a c_a} = \frac{6.64 \times 10^3}{(2.2)(140)} = \boxed{21.6\,\mathrm{K}}$$

[2]

(iv) Any two features + explanation, e.g. rotating anode (spreads heat), high thermal conductivity backing/oil cooling/forced convection, short exposure pulsing, heat-capacity optimized target materials (e.g. tungsten-rhenium). 2 marks per valid explained feature. [4]

(c) (i)

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{0.42} = 14.96\,\mathrm{rad\ s^{-1}}$$

$$v = r\omega = (0.80)(14.96) = \boxed{11.97\,\mathrm{m\ s^{-1}}}$$

So $\boxed{\omega \approx 15.0\,\mathrm{rad\ s^{-1}}}$. [3]

(ii)

$$F_c = mr\omega^2 = (55)(0.80)(14.96)^2 = \boxed{9.85 \times 10^3\,\mathrm{N}}$$

[2]

(iii) Open-ended marking (max 4):

- Image-quality consequences (any two): shorter acquisition reduces motion blur; but lower photon count per projection may increase quantum noise unless current/dose raised.
- Safety/engineering consequences (any two): increased centripetal force/vibration and mechanical stress; greater bearing wear and heat load; possible increase in patient dose if current increased to maintain SNR; tighter control requirements for slip-ring/power electronics.

1 mark per valid explained point (max 4). [4]

(d) (i) Sequential attenuation:

$$I = I_0 e^{-\mu_b x_b} e^{-\mu_s x_s} = I_0 e^{-(\mu_b x_b + \mu_s x_s)}$$

[2]

(ii)

$$\frac{I}{I_0} = e^{-[(0.62)(5.0)+(0.21)(10.0)]} = e^{-5.20} = \boxed{5.52 \times 10^{-3}}$$

(about 0.552% transmitted). [2]

(iii) Additional iodine path:

$$\frac{I'}{I_0} = e^{-[(0.62)(5.0)+(0.21)(10.0)+(0.38)(2.0)]} = e^{-5.96} = \boxed{2.58 \times 10^{-3}}$$

[2]

(iv) Any three explained points: iodine has higher effective attenuation than blood/soft tissue at diagnostic energies; this increases projection contrast; reconstruction maps larger attenuation differences to clearer voxel-value separation, improving vessel delineation. [3]

(e) (i) Use $V_{s,\text{rms}} = V_{s,\text{peak}}/\sqrt{2} = 140\,\text{kV}/\sqrt{2} = 98.99\,\text{kV}$. Then

$$\frac{N_s}{N_p} = \frac{V_{s,\text{rms}}}{V_{p,\text{rms}}} = \frac{98.99 \times 10^3}{600} = 164.98$$

$$N_s = (240)(164.98) = \boxed{3.96 \times 10^4\,\text{turns}}$$

(about $\boxed{39600}$ turns). [3]

(ii) Rectification gives unidirectional tube voltage so electrons accelerate only toward anode; smoothing reduces ripple so X-ray output and beam quality are stable, improving image consistency. [2]

(iii) One limitation + effect (any one explained): ignores winding resistance/core losses/leakage flux/saturation; therefore real secondary voltage is lower and load-dependent, requiring regulation and feedback control. [2]